

An Open-Source FORTRAN90 Code Using Vortex Rings and a Vortex Cylinder for a Constantly Loaded Actuator Disk *

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Abstract

An open-source FORTRAN90 implementation of a vortex ring/-cylinder code applied to a constantly loaded actuator disk is described. Results for $c_T = 1$ (propeller) and $c_T = -8/9$ (Betz wind turbine) are presented and compared with data from Prof. em. Gijs van Kuik [9]. Version 5 added an analysis of the two convergence measures used by the code, an improved slip-stream iteration (V3 of the code: $\sqrt{z}$ -damping and Anderson acceleration), and a review of the disc-edge singularity. Version 6 adds systematic N - and z_{SIVC} -studies with Richardson extrapolation of c_P , an analysis of the “fine-tuning” convergence scheme of van Kuik & Lignarolo, and a structurally new discretization (V4 of the code): vortex rings collocated at nodes with a spacing-proportional vortex kernel, which converges to a unique fixed point with machine-accuracy residuals. The code is available on GitHub:

<https://github.com/Schaffarczyk/RingsCylinderActuatorDisk/>

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1 Introduction

Rotors, and especially wind-turbine rotors, have been modeled by an axisymmetric actuator disk for more than 150 years, see for example [24, 14]. Originally, only the axial velocity component was used in an integral momentum-theory description [16], but it can easily be shown that this is not consistent with the differential equations of inviscid fluid flow, Euler’s equations:

$$\nabla \cdot \mathbf{u} = 0 , \tag{1}$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mathbf{f} . \tag{2}$$

Therefore, to include the radial velocity component, vortex-ring methods [3, 22] appear to be the simplest way to overcome this shortcoming if one does not want to solve the differential equations themselves numerically.

Beginning with Stig Øye’s paper from 1990 [11], this approach has been further developed and applied in [23, 1, 2]. It was generalized to non-uniform loading by Conway [8].

Here we report on an independent FORTRAN 90/95 [6] implementation of this model.

The main objectives are:

- to scrutinize the accuracy of the model,
- to understand what the two convergence measures of the code (the sheet residual and the deviation of c_P from momentum theory) do and do not measure (new in Version 5, Sec. 2.4),
- to look for hints of the properties of a possible edge singularity [18, 7] (Sec. 7).

2 Implementation

2.1 Model description

We use axisymmetric coordinates (z, r) with velocity $\mathbf{u} = (u_z, u_r)$. The actuator disk is located at $z = 0$ and $0 \leq r < 1$. The edge $(\mathbf{r_E} = (0, 1))$ may carry a singularity, i.e. a point where $\mathbf{u}$ is not defined.

The model itself is defined by a piecewise linear vortex sheet which starts at $\mathbf{r}_{\text{edge}} = (0, 1)$ and ends at $\mathbf{r}_{\text{Cyl}} = (z_{SIVC}, r_{SIVC})$, the location where a semi-infinite vortex cylinder is appended. For $0 < z < z_{SIVC}$, NP vortex rings with strength γ are placed with non-uniform spacing at locations (z^{sl}, r^{sl}) . This part of our approach is identical to that of [1, 2]. The axial positions of the rings are distributed according to

$$z_i = z_{SIVC} (1 - \cos \varphi_i), \quad (3)$$

$$\varphi_i = \frac{\pi}{2} \cdot \frac{i}{NP}, \quad 1 \leq i \leq NP. \quad (4)$$

To circumvent singularities on the slipstream (i.e. infinite velocities), all velocities are evaluated at the (arithmetic) midpoint of two neighboring slipstream (singularity-carrying) points. This approach avoids introducing an artificial *vortex core*.

The basic model parameter is the thrust coefficient c_T , where $c_T > 0$ refers to the propeller case (induced velocities > 0) and $c_T < 0$ to the turbine case. Momentum theory is valid for $-1 \leq c_T < \infty$. Many parameters can be deduced from momentum theory, for example the asymptotic radial extension of the slip-stream, which is given by

$$r_{z \rightarrow \infty}^{sl} \equiv r_{SIVC} = \sqrt{\frac{1 + \sqrt{1 + c_T}}{2 \cdot \sqrt{1 + c_T}}}, \quad (5)$$

$$\text{for } c_T = 1 \text{ we have } r_\infty = 0.9239 \text{ and} \quad (6)$$

$$\text{for } c_T = -8/9: r_\infty = \sqrt{2} = 1.4142. \quad (7)$$

The value of the (total, i.e. inflow included) stream function on the slipstream far downstream is

$$\psi_{wake} = \frac{1}{4} (1 + \sqrt{1 + c_T}). \quad (8)$$

For $c_T = 1$ this gives $\psi_{wake} = 0.603553$, for $c_T = -8/9$: $\psi_{wake} = 1/3$. (If only the induced part of the stream function is used, the corresponding value is $c_T/(4\sqrt{1 + c_T})$; the code works with the total stream function, Eq. (8).)

The parameters defining our model are summarized in Table 1. The inflow velocity is set to $u_z(r)_{z \rightarrow -\infty} = 1.0$.

2.2 Induced Velocities

Accurate calculation of the induced axial and radial velocities is the key to any arbitrarily accurate (= successful) model. The calculation is divided into

name	meaning
major	
c_T	thrust coefficient
restart	LOGICAL; if .T. the code restarts from an existing sls.DAT
NP	number of rings
ZSIVC	axial location of the semi-infinite cylinder
minor	
selfind	(0,1), switch self-induction off/on
update	(0,1), default = 1, debugging only
maxiter	maximum number of slip-stream iterations (≤ 2000 used)
under	under-relaxation parameter (see Sec. 2.4)
epsSL	stop tolerance on the sheet residual errsl
wakety	initial wake shape (3 models available)
wakety=1	$rsl_i = r_\infty + (1 - r_\infty) \exp(-a \cdot zsl_i)$
wakety=2	gnuplot fit of vK data, $c_T = 1$ and $c_T = -8/9$ only
wakety=3	van Kuik's Eq. (D.10)
wake-a	parameter a for wakety = 1
iter-sc	slip-stream iteration scheme (3 = ψ ; 7 = ψ with $\sqrt{z}$ -damping; 8 = 7 + Anderson; V3)

Table 1: Major and minor model parameters

three parts:

1. induced velocities from the $NP - 1$ other vortex rings onto the ring under consideration, for $0 \leq z \leq Z_{SIVC}$,
2. self-induced velocities of the ring under consideration:
we have implemented only Bontempo's [2] Eqs. (6) and (7), which, surprisingly, do not contain any core parameter:

$$v_{z,m \rightarrow m} = \gamma_m \left\{ -\frac{\beta_{m+1} - \beta_m}{8\pi} \cos \beta_m - \frac{ds_m}{4\pi r_m} \left[\log \left(\frac{8\pi r_m}{ds_m} \right) - 0.25 \right] \right\} \quad (9)$$

$$v_{r,m \rightarrow m} = \gamma_m \left\{ \sin(\beta_m) - \frac{\beta_{m+1} - \beta_m}{8\pi} \right\} \quad (10)$$

with β_m the slope at ring m ,
 r_m the radial position of ring m and
 ds_m the distance between the two neighboring rings.

For more details on the self-induction of vortex rings see, for example, [13].

3. induced velocities from a semi-infinite vortex cylinder with tangential vorticity located at Z_{SIVC} [3], chapter 36.

To avoid implementation errors, three different *models* (i.e. sets of equations) can be chosen:

1. Eqs. (1) to (7) from [1, 2], called model *Bontempo*,
2. Eqs. (D1) from [22], called model *van Kuik*, and
3. equations from chapters 33 and 36 of [3], called model *Branlard*.

To keep the nomenclature as consistent as possible, we prefer (and use) equations from Branlard [3] throughout.

2.3 Slip-Stream Iteration

The shape of the slipstream is determined by the induced velocities, and vice versa, through the constraints described in [1, 2, 22]:

- the force-free condition, implemented as $\gamma = -c_T/(2 \cdot u_s)$ with $u_s^2 = (1 + u_z)^2 + u_r^2$,
- the slip stream aligned with the flow, implemented as $\psi_w(z) = \psi_w(\infty)$.

An initial slip-stream shape may be helpful to reduce the number of iterations. The following models are implemented:

1. a cylinder with $r_{SIVC} = 1$, or
2. a prescribed exponential shape described by

$$r_{sl}(z) = r_{SIVC} + (r_0 - r_{SIVC}) \exp(-a \cdot z), \quad (11)$$

3. a prescribed algebraic shape deduced from the axial velocity at $r = 0$ of a vortex cylinder (attributed by van Kuik [22] to Snel and Schepers):

$$u_z(r = 0, z) = \frac{\gamma_\infty}{2} \left(1 + \frac{z}{\sqrt{z^2 + r_{SIVC}^2}} \right). \quad (12)$$

r_{SIVC} is given by Eq. (7), but a (not to be confused with the axial induction factor) has to be guessed, $a \approx 1 \dots 2$. Fig. 1 shows a GNUPLOT fit of van Kuik's $c_T = 1$ data. As he sets $r_{SIVC,vK} := 1$, a fit results in:

1. $r_0 = 1.03498$,
2. $r_{SIVC} = 1.00006$ (as it should be),
3. $a = 1.01452$.

We have implemented several algorithms for the slip-stream iteration (input parameter `iter-sc`, cf. Table 1):

1. Eqs. (13) and (14), Fig. 2 from [2],
2. shifting points normal to the current slip-stream direction, or
3. van Kuik's [22] Eq. (D.12),

$$\boxed{\partial r(z) = \frac{\partial \psi}{r \cdot u_z}}$$

with $\partial \psi = \psi(\text{slip-stream}, z) - \psi_{wake}$ from Eq. (8);

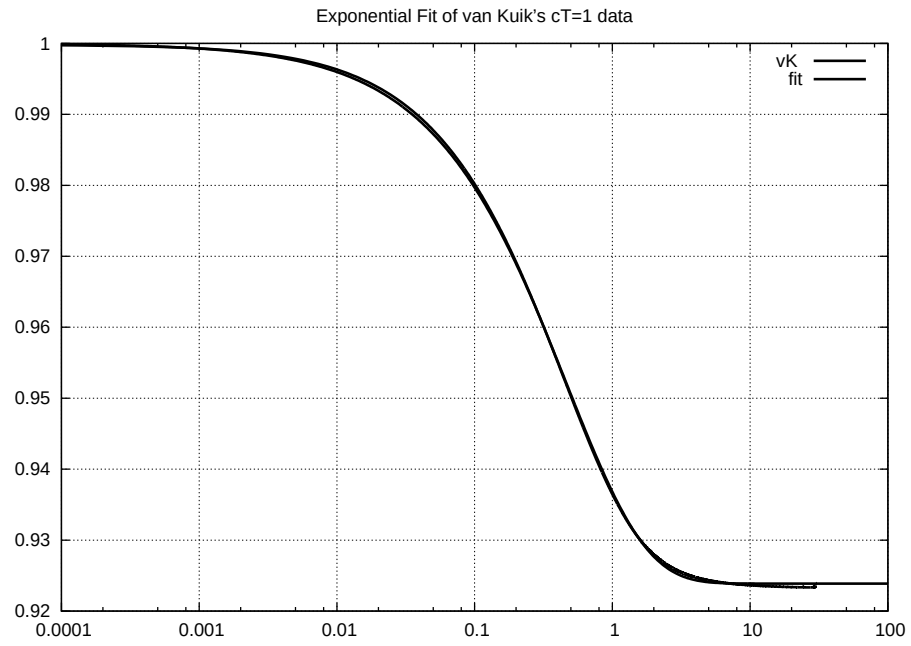


Figure 1: Two-parameter (a_2, k) fit of van Kuik's $c_T = 1$ slip stream according to $r(z) = r_\infty + (1 - r_\infty) \exp(-a_2 z^k)$. Here: $a_2 = 1.7743$ and $k = 0.7570$.

7. as (3), but with the damping factor $\lambda(z) = \min(0.1, \textit{under} \cdot \sqrt{z})$ instead of $\min(z, 0.1) \cdot \textit{under}$ (scheme 7, code version V3, 2026; see Sec. 2.4);
8. as (7), plus Anderson acceleration (window 4) of the resulting fixed-point iteration (scheme 8, V3).

Schemes 7 and 8 are to be preferred; among the older schemes, No. 3 was the preferred one. All algorithms may be under-relaxed. The corresponding factors have to be chosen:

1. $0.001 \cdots 0.01$,
2. < 0.01 ,
3. $0.1 \cdots 0.9$,
- 7./8. $\textit{under} = 0.05 \cdots 0.1$ (acts as the prefactor of $\sqrt{z}$).

The slip-stream iteration ends if

1. the number of iterations exceeds MAXITER, or
2. the sheet residual satisfies $\text{errsl} \leq \text{epsSL}$, or
3. (schemes 7/8) c_P has stopped moving: $|\Delta c_P|$ over the last 100 iterations $< 10^{-6}$ (Cauchy criterion).

Note that in earlier versions the iteration was stopped when $\text{errc}_P := |c_P - c_{P,mom}|$ fell below a tolerance. As shown in Sec. 2.4, errc_P is **not** a valid iteration criterion: it possesses a discretization floor, and small values can also be produced by mere zero crossings of $c_P - c_{P,mom}$ during the iteration.

2.4 Convergence measures, stability and the improved iteration (V3)

This section summarizes an analysis carried out in 2026 with the help of a 1:1 python replica of the code (numpy/scipy, same Carlson elliptic integrals), which allowed systematic parameter studies ($NP = 300 \dots 2400$, up to 15 000 iterations).

2.4.1 What errsl and errc_P measure

The code monitors two quantities:

$$\text{errsl} = \frac{1}{NP} \sum_{i=1}^{NP} |\psi(z_i, r_i) - \psi_{wake}| ds_i, \quad (13)$$

$$\text{errc}_P = \left| \frac{c_P - c_{P,mom}}{c_{P,mom}} \right|. \quad (14)$$

These measure two different things: errsl is the residual of the *discrete* system, whereas errc_P is the distance of the discrete solution to the *continuum* momentum-theory limit. Even at the exact discrete fixed point ($\text{errsl} = 0$), errc_P retains a discretization floor: $\sim 1/NP^2$ from the ring discretization (Table 6), plus a truncation error from the finite z_{SIVC} , plus the contribution of the disc-edge region, which midpoint panels cannot resolve (Sec. 7). Hence $\text{errsl} < 10^{-8}$ does not, and cannot, imply $\text{errc}_P < 10^{-8}$.

Three further mechanisms de-couple the two measures:

1. **The ds -weighting makes errsl blind at the disc edge.** With the cosine clustering, $ds \approx z_{SIVC} \pi^2 / (4NP^2)$ at the first ring — precisely where $|\Delta\psi|$ is largest (edge singularity) and where c_P is most sensitive (the weight $2r dr$ of the power integration concentrates near $r = 1$). Since V3, the code therefore also prints $\max |\Delta\psi|$ (unweighted).
2. **The damping factor $\min(z, 0.1)$ of scheme 3 freezes the edge.** The relaxation factor $\lambda(z) \propto z$ vanishes at the disc, so the equilibration time of the first ring is $\sim 1/z_1 \approx 10^4$ iterations ($NP = 600$, $z_{SIVC} = 25$): errsl reaches its floor after ≈ 200 iterations while c_P keeps creeping for thousands of iterations. A 15 000-iteration reference run never reached a fixed point: c_P wandered by $\pm 2 \cdot 10^{-3}$ around $c_{P,mom}$ at $\text{errsl} \approx 10^{-9}$.
3. **errc_P is not monotone.** During this creep, c_P crosses $c_{P,mom}$; very small errc_P values (down to 10^{-6}) occur at these zero crossings without indicating convergence.

Two implementation issues were found and corrected in V2/V3 of the code: the segment length ds used in errsl was taken from a stale variable (making $\text{errsl} \approx 50$ times too large), and the under-relaxation factor was applied

twice (effective relaxation $\min(z, 0.1) \cdot \text{under}^2$). In addition, the module file `mem.f` must be compiled with `-fdefault-real-8` as well; otherwise the global arrays remain 32-bit while the rest of the code works in 64-bit.

2.4.2 Stability: the saw-tooth (Kelvin–Helmholtz) mode

A cylindrical vortex sheet is Kelvin–Helmholtz unstable; the discrete manifestation is a saw-tooth mode on the grid scale. The damping of scheme 3 suppresses it — and at the same time freezes the edge. Systematic tests (replica, $NP = 600$) gave:

- undamped Newton steps $\partial r = \partial\psi/(ru_z)$: the saw-tooth amplitude grows from 10^{-2} to $O(1)$; c_P diverges;
- constant damping: marginally stable at $\lambda \approx 0.003$ ($NP = 600$), unstable for larger λ ; the threshold *decreases* with NP ;
- $\lambda \propto z$ (scheme 3): stable but edge frozen;
- $\lambda = \min(0.1, \text{under}\sqrt{z})$ (scheme 7): stable for $NP = 600$ and 1200, edge $\approx 100\times$ faster than scheme 3.

Purely local alternatives (per-station Newton with a geometry-consistent finite-difference Jacobian, Gauss–Seidel sweeps, naive streamline tracing) were also tested and either diverge or converge to a shifted fixed point; the ring-to-ring coupling near the edge is too strong for any purely local update.

2.4.3 The improved iteration (schemes 7 and 8) and the c_P band

With scheme 7, errc_P drops to $\approx 10^{-4}$ within ≈ 350 iterations ($NP = 600$, $z_{SIVC} = 25$) and then oscillates slowly around $c_{P,mom}$; with the Anderson-accelerated scheme 8, errsl reaches its floor 3–4 $\times$ faster still. The remaining oscillation band of c_P , $\pm(3\dots 5) \cdot 10^{-4}$ for $NP = 600$, $z_{SIVC} = 25$, is **not** an iteration deficiency: at $\text{errsl} \approx 10^{-8}$ the discrete equations leave the unresolved edge region — and with it c_P — undetermined within this band. Different (all stable) update schemes converge to slightly different pseudo-solutions inside the same band. The band shrinks with increasing NP and z_{SIVC} , not with more iterations.

Since V3 the code therefore reports, in addition to the last iterate, an exponentially weighted moving average of c_P (≈ 50 iterations) and the fluctuation band $\pm(\max - \min)/2$ of c_P over the last 100 iterations. We recommend

N	$\bar{c}_P$	band $\pm$	$\bar{c}_P - c_{P,mom}$
1000	-0.5923884	$3.8 \cdot 10^{-4}$	$+2.0 \cdot 10^{-4}$
2000	-0.5924765	$6.8 \cdot 10^{-5}$	$+1.2 \cdot 10^{-4}$
4000	-0.5925145	$5.3 \cdot 10^{-5}$	$+7.8 \cdot 10^{-5}$

Table 2: N -study, Betz case, $z_{SIVC} = 50$, scheme 7 (V3).

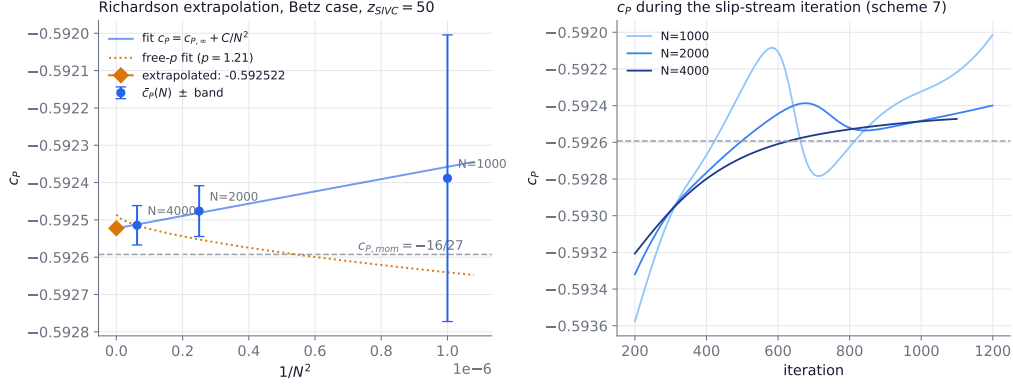


Figure 2: Richardson extrapolation of $c_P(N)$ for the V3 scheme (left) and the evolution of c_P during the iteration (right), Betz case, $z_{SIVC} = 50$.

quoting c_P as “average $\pm$ band” and reducing the band by an NP -study with Richardson extrapolation in $1/NP^2$.

2.4.4 N - and z_{SIVC} -studies with Richardson extrapolation (V6)

The Betz case was computed with scheme 7 for $N = 1000/2000/4000$ at $z_{SIVC} = 50$ (1100–1200 iterations each; python replica of the code). The tail means and fluctuation bands are given in Table 2; Fig. 2 shows the Richardson extrapolation. All extrapolation variants land at $c_{P,\infty} = -0.59250 \dots - 0.59253$, i.e. within $(0.6 \dots 1.1) \cdot 10^{-4}$ of $-16/27$; the observed convergence order is $p \approx 1.2$ — edge-dominated, not the smooth $1/N^2$ of the ring discretization. The recommended citation is $c_P = -0.59251 \pm 0.00007$ for this protocol.

A z_{SIVC} -study at fixed $N = 2000$ ($z_{SIVC} = 50/100/200$) gave the perhaps counter-intuitive result that **larger z_{SIVC} degrades the result**: the bands grow from $6.8 \cdot 10^{-5}$ to $2.0 \cdot 10^{-4}$ and $3.9 \cdot 10^{-4}$, and the means move away

from $-16/27$. The reason is that at fixed N all cell sizes scale $\propto z_{SIVC}$ (the edge cell even $\propto z_{SIVC}/N^2$), so the loss of resolution outweighs the smaller truncation error of the cylinder, which is already subdominant at $z_{SIVC} = 50$. Practical rule: use $z_{SIVC} = 50$ and invest in N ; a double extrapolation in (N, z_{SIVC}) is unnecessary.

3 Numerics

3.1 The goal of numerical computations

Everybody knows from their own experience the NEWTON–(Raphson) method for finding zeros of functions ($f(x_0) = 0$) or SIMPSON’s rule for 1D integration. The ultimate manifestation of numerical algorithms may be found in the famous (series of) textbook(s): *NUMERICAL RECIPES in Fortran 77* [12].

Let us state

Theorem 1. *A numerical scheme is meaningful if and only if*

$$\forall \epsilon > 0 \exists n \in \mathbb{N}$$

such that after n iterations the result is accurate to ϵ .

For the sheet residual errsl, version V3 of the code demonstrably satisfies this theorem. For errc_P it can hold only in the combined limit (iterations $\rightarrow \infty$, $NP \rightarrow \infty$, $z_{SIVC} \rightarrow \infty$), because of the discretization floor discussed in Sec. 2.4.

3.2 A short note on FORTRAN [6]

We will not defend why we use the *grandfather of all (higher) programming languages*, but simply use it. We have tried to write the code as simply as possible, to make understanding and changes (**i.e. maintainability**) as easy as possible. A minimal set of algorithmic features is used. In particular, no tricks from computer science have been applied.

3.2.1 Compilation

For compilation we use (in a LINUX environment) *GNU Fortran* (in short: gfortran) [10] with the mandatory options (see **compVortexCode.cmd**):

- fno-automatic helps to resolve notorious issues with local variables in SUBROUTINES and FUNCTIONS,
- fdefault-real-8 sets all floating-point numbers to REAL*8 (= 64 bit). The 53-bit significand corresponds to ≈ 15 –16 decimal digits, with a range of about $\pm 10^{\pm 308}$. NOTE (V3): this option must be applied to **all** source files including mem.f; otherwise the global arrays of the module remain 32-bit.

3.2.2 Modularization

The code is divided into only three parts:

- mem.f, declaring global variables,
- VortexCode.f, the main slip-stream iteration loop,
- SubA.f, the set of functions used, including the evaluation of elliptic integrals (from an external source).

3.3 Elliptic Integrals

The elliptic integrals, see chapter (C.4) of [3], are evaluated via the open-source subroutine **RFL** of [5]:

```
c***begin prologue  rfl
c***date  written    790801    (yymmdd)
c***revision  date   831228    (yymmdd)
c***category      no.   c14
c***keywords  duplication theorem, elliptic integral, incomplete, complete
c              integral of the first kind, taylor series
c***author    carlson, b.c., ames laboratory—doe
c              iowa state university, ames, iowa  50011
c              notis, e.m., ames laboratory—doe
c              iowa state university, ames, iowa  50011
c              pexton, r.l., lawrence livermore national laboratory
c              livermore, california  94550
```

```

restart      . false .
useVC        . true .
N-Rings      2500
ct           -0.8888888888
zsivc        50
selfind      . true .
update       . true .
maxiter      2000
under        0.1
epsSL        1.e-8
IndVel       3
iniwake      3
wake-a       1.0
SLiter       7

```

Table 3: Sample input (configuration) file

It was tested and agrees with MATHEMATICA to 10 digits.

3.4 Evaluation of induced velocities

The accuracy of the induced velocities was tested separately in the following manner, using all cases from list 3:

- check $v_{z,r}$ for the ring part $0 < z < z_{SIVC}$,
- check $v_{z,r}$ for the cylinder part $z \geq z_{SIVC}$,
- compare with data from van Kuik [9].

3.5 Verification and Evaluation

The power is determined using Bontempo's Eqs. (15) to (20) [2]. The radial integration of $u_z(z = 0, r)$ from $r = 0$ to $r = 1$ is performed by Simpson's rule. This scheme is second-order accurate; thus, for an accuracy of $< 10^{-6}$, at least 10^3 evaluation points are needed. We use 3000 points.

A computational case is defined in **inpa.dat** by the following set-up parameters (V3 sample):

Before testing the iteration scheme, two types of fixed slipstream shapes can be examined:

1. a fixed cylinder with $r = 1$ or $r = r_{SIVC}$,
2. the prescribed shape of Eq. (11), see Fig. 1 (for $c_T = 1$ and $c_T = -8/9$ only),
3. an algebraic shape according to Eq. (12).

4 Output Files, Tools for Visualization and Testing

4.1 Description of output files

The following (ASCII) output is provided (list not complete):

1. sls.DAT: geometry, velocity, vortex strength and stream function on the vortex sheet,
2. vzd.DAT, vrd.DAT: velocities at the disk ($z = 0$),
3. vza.DAT: axial velocity on the axis ($r = 0$),
4. psid.DAT: stream function at the disk ($z = 0$),
5. conv.DAT: iteration history; since V3 with the additional columns $\max |\Delta\psi|$ and $|\Delta c_P|$ over the last 100 iterations,
6. streaml.DAT: 3D data for contour plots of streamlines (by str-cont.py).

4.2 Description of GNUPLOT scripts

1. SLS-vz-(Prop,WT).gpl and SLS-vr-(Prop,WT).gpl: plot shape and velocities of the slip stream,
2. vza.gpl and vzd.gpl: plot axial and radial velocity at the disk,
3. gamma.gpl: plot vortex strength,
4. and more.

Some GNUPLOT scripts (waked.gpl and others) have to be executed via
`gnuplot -e "ct=1.0; load 'waked.gpl'"`
to allow for input of c_T .

4.3 One Python script

1. str-cont.py: plots contour lines of the stream function together with the slipstream (red).

4.4 Run-time estimation

As expected, the CPU time scales approximately $\sim NP^2$, see Fig. 3.

5 V4: node collocation with a vortex kernel (V6)

The findings of Sections 2.4 and 7 identify a structural conflict of the midpoint-collocation scheme: stability requires a position-dependent damping, the edge region remains under-determined, and c_P retains a fluctuation band. Version 4 of the code (directory OwnCode-V4) therefore changes the discretization, following van Kuik & Lignarolo [23] (Appendix A+B):

1. N vortex rings with circulations Γ_i are collocated **at the nodes** (x_i, r_i) ; ring 1 is **fixed at the disc edge** $(0, 1)$ and exempted from the boundary conditions (cf. the discussion of the leading edge in [23], Sec. 4.3).
2. A **vortex kernel** regularizes all evaluations at distances $\rho < \frac{1}{2}\beta ds_i$ (smoothed, Marshall-type expressions). Deviating from the fixed $\delta = 0.002$ of [23], the kernel radius is **proportional to the local spacing**, $\delta_i = \beta ds_i$ with $\beta = e^{-1/2} \approx 0.607$, which matches the analytic contribution of the local sheet strip. (A fixed δ biases c_P by $\sim 10^{-2}$ as soon as $ds \gg \delta$ varies along the sheet — measured with the prototype.)
3. The convergence scheme is **single-phase with uniform damping** $d = 0.05$: $\Gamma_i \leftarrow \Gamma_i + d(\Gamma_i^* - \Gamma_i)$ with $\Gamma_i^* = -c_T/(2v_i) ds_i$, and $r_i \leftarrow r_i + d(\psi_{wake} - \psi_i)/(r_i(1 + v_{z,i}))$. The kernel removes the saw-tooth mode, so no z -dependent damping is needed. The v_n “fine-tuning” phase of [23] (App. B.3) was tested and found to be neither stable nor

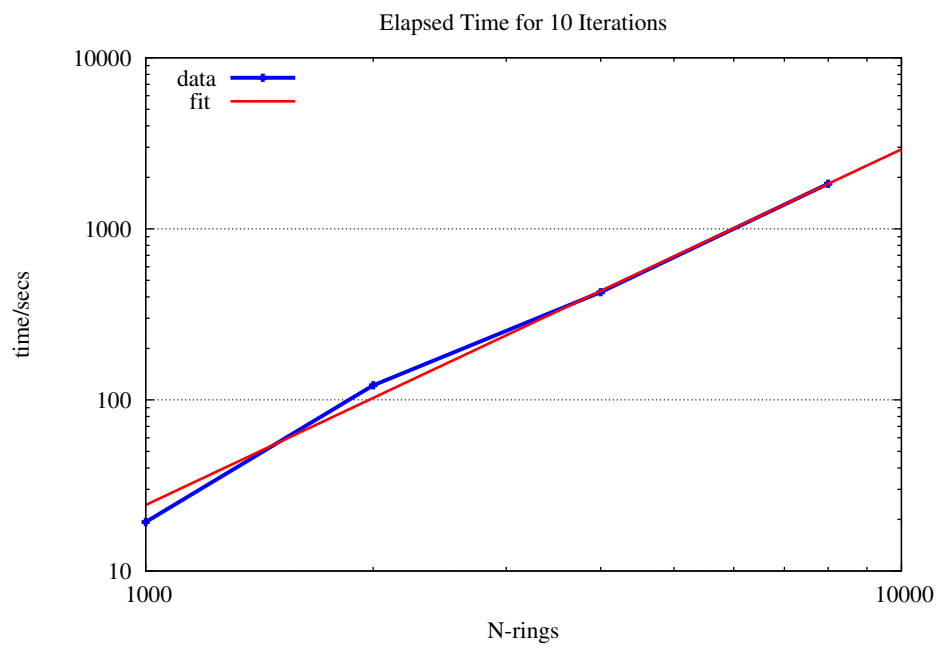


Figure 3: Elapsed CPU time for 10 slip-stream iterations. Hardware: Intel i3-4160 at 3.6 GHz. Operating system: openSUSE Leap 42.3 Linux, kernel 4.12.14-lp151.28.44-default

necessary in this implementation: after the ψ -phase, v_n is $\sim 3 \cdot 10^{-2}$ at the edge and $\sim 2 \cdot 10^{-3}$ on the sheet; it is retained as a diagnostic.

Validation (Betz case; python prototype, identical algorithm):

N	$\max \Delta\psi $	c_P	$c_P - (-16/27)$
400	$4 \cdot 10^{-12}$	-0.5920118	$+5.8 \cdot 10^{-4}$
800	$6 \cdot 10^{-12}$	-0.5923623	$+2.3 \cdot 10^{-4}$
1600	$4 \cdot 10^{-12}$	-0.5925434	$+4.9 \cdot 10^{-5}$
3200	$9 \cdot 10^{-12}$	-0.5925454	$+4.7 \cdot 10^{-5}$

Table 4: V4 validation: the iteration converges to a *unique* fixed point (residuals at machine level, no c_P band, no drift).

The error saturates between $N = 1600$ and 3200 : the V4 fixed points converge (Aitken limit) to $c_{P,lim} = -0.592545$, i.e. a remaining, N -independent **model bias** of $+4.7 \cdot 10^{-5}$ with respect to momentum theory, while the pre-saturation discretization error decays with observed order $p \approx 1$ (Fig. 4). Separating the origin of this bias (tube truncation at $L_{tube} = 30R_{wake}$ vs. the kernel model at the junction and edge) requires runs with $N \propto L_{tube}$; at fixed N , enlarging L_{tube} degrades the result because all cells coarsen — the same resolution/truncation confounding as in the z_{SIVC} -study of Sec. 2.4.4.

In contrast to V1–V3, the remaining c_P error of V4 is a smooth, pure discretization error: there is no band and no drift, and the Richardson extrapolation is correspondingly clean (Fig. 4). The edge singularity itself remains kernel-smoothed; resolving the conjectured $1/\sqrt{s}$ zone (Sec. 7) would still require an analytic edge cell or a multivalued sheet parametrization.

6 Results for sample cases

6.1 Sample case 1: propeller with $c_T = 1$

As can be seen from the numbers above, this case is only moderately non-linear. The slip-stream contraction is down to approx. 0.93 and $c_P = 1.207$, compared to a linear approximation [4], Eq. (9.107), which gives $c_{P,lin} = 1 + c_T/4 = 1.25$.

Some sample figures show the comparison of:

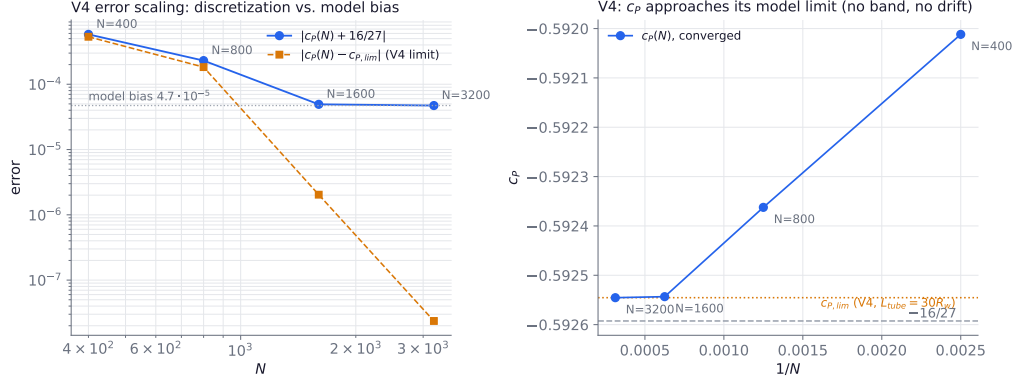


Figure 4: V4 N -study, Betz case: error scaling of the converged fixed points (left; circles: distance to $-16/27$, squares: distance to the V4 limit) and $c_P(N)$ with the V4 model limit (right).

1. the induced axial velocity on the slip stream, Fig. 5,
 2. the vortex strength on the slip stream, Fig. 6,
 3. streamlines and slip stream, Fig. 7
- with van Kuik's [9] data.

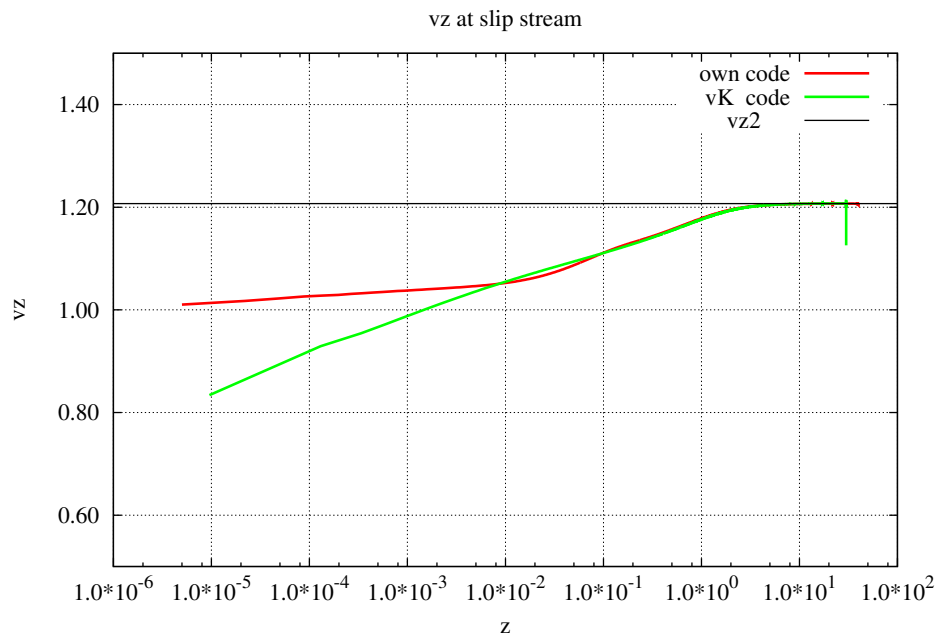


Figure 5: Calculated axial distribution of the induced axial velocity, $c_T = 1$.

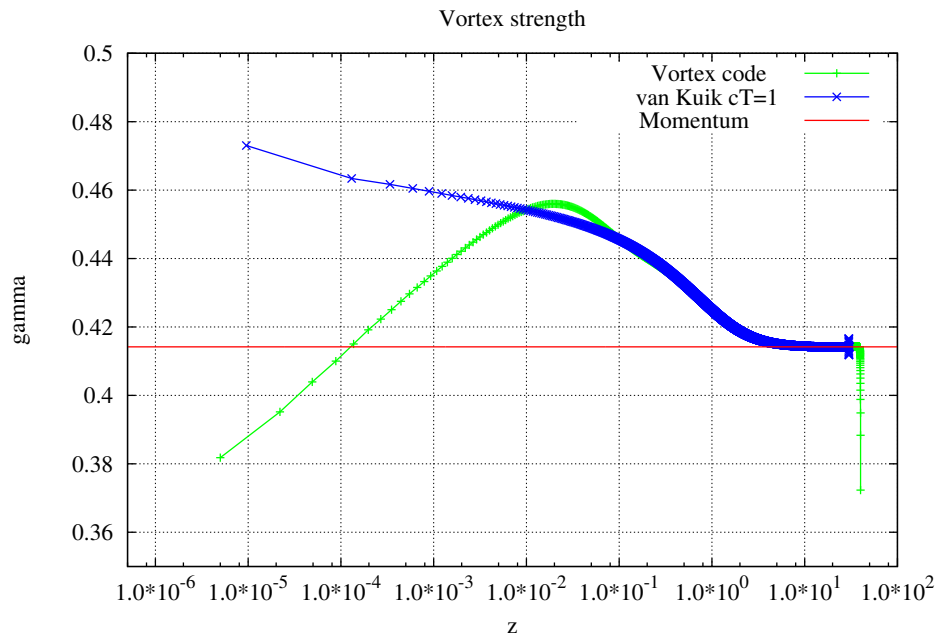


Figure 6: Calculated axial distribution of the vortex strength, $c_T = 1$.

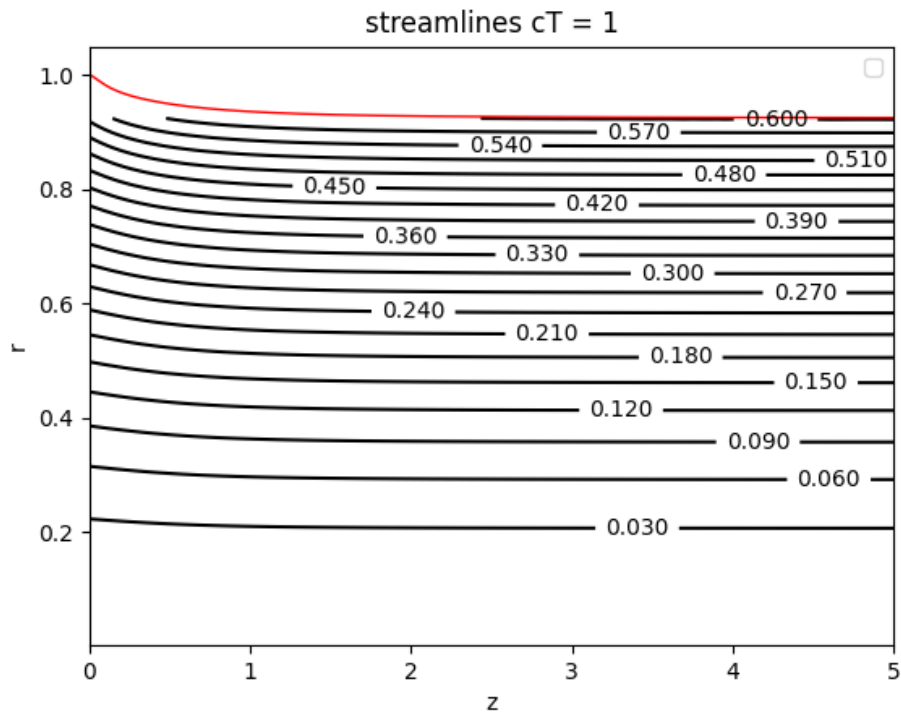


Figure 7: Calculated streamlines (lines of $\Psi = \text{const.}$) for $c_T = 1$. In red: wake boundary. As can be seen, the edge streamline coincides well with $\Psi_{wake} = 0.60355339$.

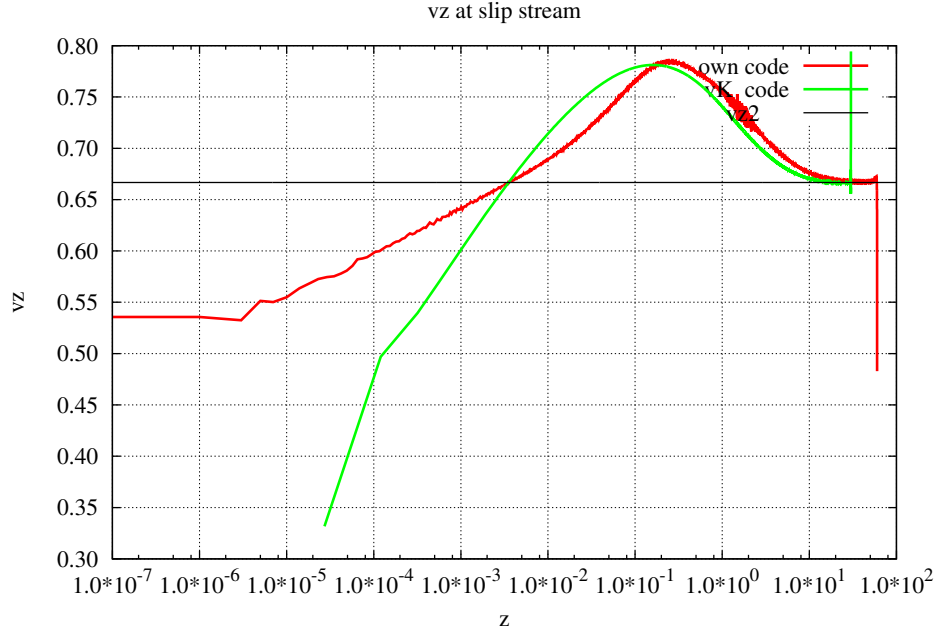


Figure 8: Calculated axial distribution of the induced axial velocity, Betz case

6.2 Sample case 2: (Betz) turbine with $c_T = -8/9$

Same plots (with one addition) as shown in subsection 6.1:

1. induced axial velocity on the slip stream, Fig. 8,
2. induced radial velocity on the slip stream, Fig. 9,
3. vortex strength on the slip stream, Fig. 10,
4. streamlines and slip stream, Fig. 11.

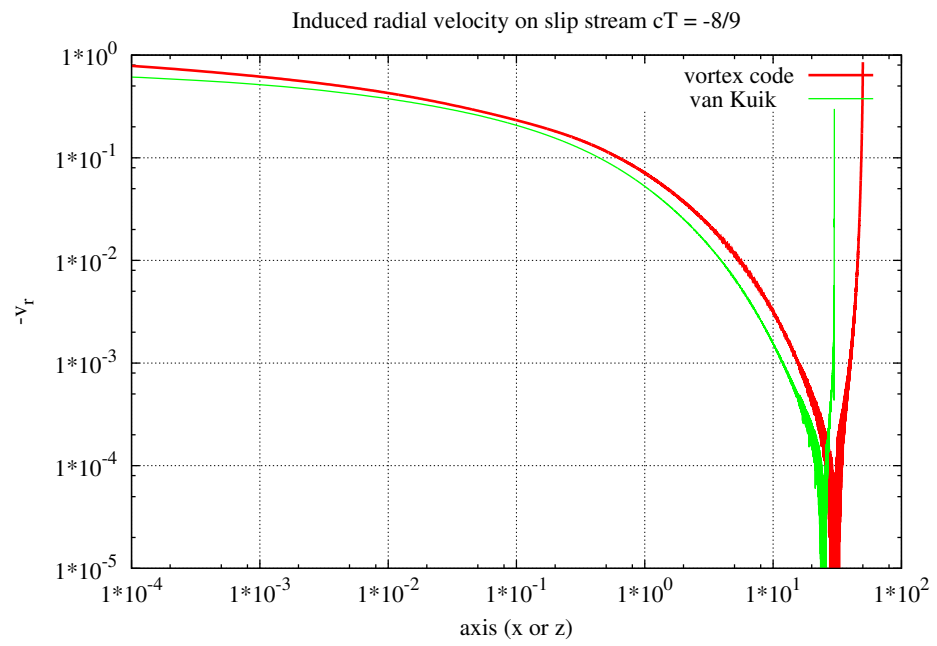


Figure 9: Calculated axial distribution of the induced radial velocity, Betz case

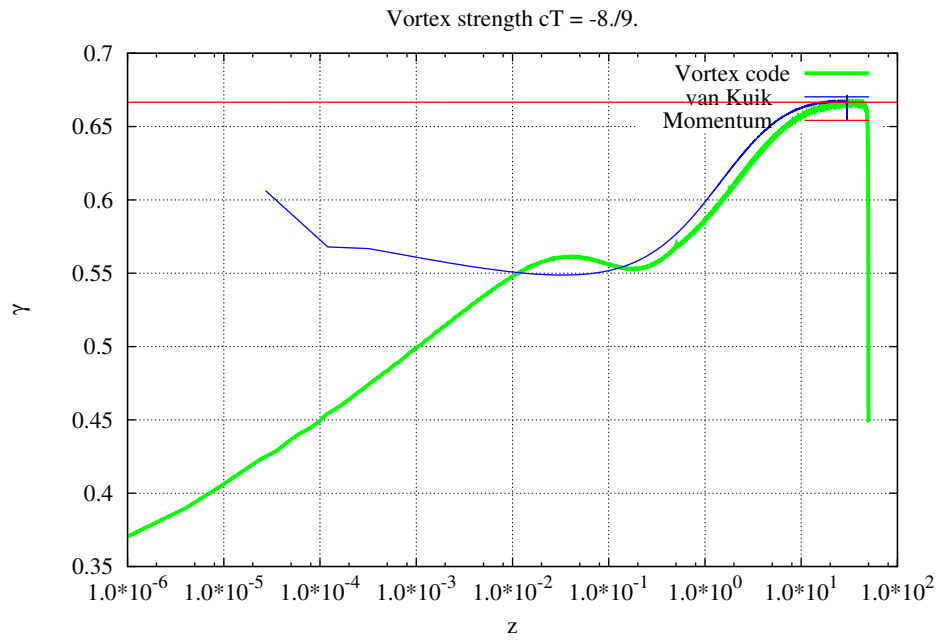


Figure 10: Axial distribution of the vortex strength, Betz case.

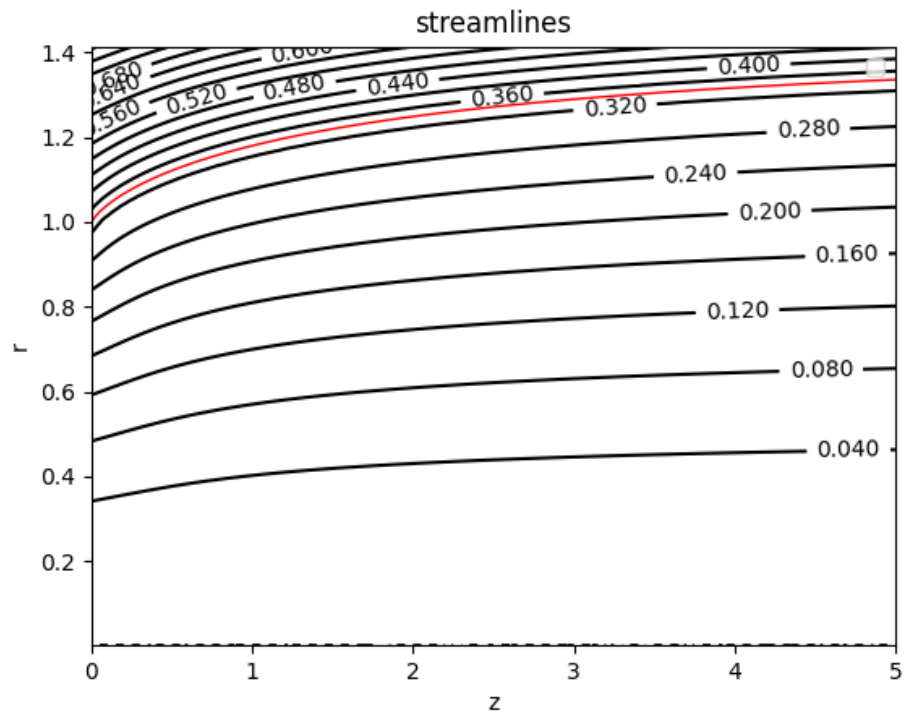


Figure 11: Streamlines for the Betz case, $c_T = -8/9$; red: slip stream, $\Psi_{wake} = 0.33333$

7 The disc-edge singularity: literature status and numerical evidence

Along a force-free vortex sheet, $\gamma u_s = -c_T/2$ is constant. Van Kuik [18, 22] proposes that at the disc edge the wake boundary rolls up into a self-similar (Prandtl/Kaden-type) spiral whose unrolled strength diverges as

$$\gamma(s) \propto s^{-1/2}, \quad s \rightarrow 0, \quad (15)$$

with s the arc length measured from the edge. Is Eq. (15) an established result or a hypothesis? A literature review together with our own numerical evidence gives the following picture.

7.1 Established results

(a) The linear (frozen-wake) model is singular at the rim. For the semi-infinite vortex cylinder of *constant* strength, the induced velocity field is singular at the rim: v_z diverges logarithmically and $\partial v_r / \partial s \rightarrow \infty$. This is analytic (visible in the very Branlard formulas used in this code), and Chattot [7] derives via Biot–Savart that the linearized slip line behaves as $r - R \propto -x \ln x$ near the edge.

(b) The non-linear force-free problem is singular at the leading edge of the sheet. All potential-flow computations that let γ adjust to the force-free condition find the sheet strength growing (resolution-limited) towards the leading edge. Van Kuik [21] states: “In all flow cases the distributions have a singularity at the leading edge”, with a load-dependent strength; see also [23]. The qualitative statement “the disc edge is singular” is therefore established.

7.2 The $1/\sqrt{s}$ spiral: strongly supported, not proven

The analytical basis is the local non-linear analysis of Schmidt & Sparenberg [15] for large constant normal load. Van Kuik revisited the edge in [18] and constructed the flow induced by, and a steady solution for, Prandtl’s self-similar vortex-sheet spirals [19, 20]; Eq. (15) is exactly the Kaden roll-up law, rigorously established for the *unsteady* two-dimensional similarity problem (see, e.g., [13]). Independent support comes from Spalart [17], who concludes — explicitly labeled a conjecture — that “the cross-section of the

fit window (rings)	s -range	a ($NP = 600$)	a ($NP = 1200$)
2–20	$5 \cdot 10^{-4} \dots 4 \cdot 10^{-2}$	−0.04	+0.02
3–40	$8 \cdot 10^{-4} \dots 1.5 \cdot 10^{-1}$	−0.03	−0.02
5–80	$1 \cdot 10^{-3} \dots 6 \cdot 10^{-1}$	+0.01	−0.02

Table 5: Fitted edge exponents a of $\gamma \propto s^a$ (Betz case, $z_{SIVC} = 25$, scheme 7, converged).

vortex sheet probably begins with a 45° spiral”, supported by a viscous numerical solution. In contrast, Chattot [7] obtains, from a linearized analysis with *prescribed constant* sheet strength, logarithmic velocity singularities and a bounded γ ; the disagreement is traceable to that assumption, since it is precisely the force-free coupling $\gamma = -c_T/(2u_s)$ that drives $\gamma \rightarrow \infty$ where $u_s \rightarrow 0$. No free-wake computation has resolved the spiral, and no experiment resolves the exponent. Thus: the *existence* of the singularity is established; the *specific form* (15) is a well-founded, independently supported, but unproven hypothesis.

7.3 Numerical evidence from this code

The converged solutions of our model show a **bounded** edge strength. A log–log fit $\gamma \propto s^a$ over the first rings of converged double-precision solutions (scheme 7) gives the exponents of Table 5: $a = 0.00 \pm 0.04$ over two decades in s , with $\gamma_{edge} \approx 0.55 < \gamma_\infty = 2/3$ and, at matched s , NP -independent values. This is consistent with the logarithmic-decrease observation of Fig. 10 and does *not* contradict the spiral picture: a discretization of the present class is structurally unable to represent it, since (i) the single-valued parametrization $r_{sl}(z)$ of the sheet excludes any multivalued (rolled-up) geometry, (ii) the self-induction term regularizes the kernel at the panel scale ds , and (iii) the force-free update with the regularized velocity keeps γ bounded. The code converges to the regularized solution of the no-roll-up problem — which is why our findings differ from van Kuik’s $s^{-1/2}$ *and* from Chattot’s constant- γ picture, as already noted in earlier versions of this report. Practically, the singularity is integrable ($\int \gamma ds < \infty$), so c_P converges unaffected; its footprint is the saturation of $\max |\Delta\psi|$ at the first ring(s) and the c_P band of Sec. 2.4.

7.4 The “fine-tuning” of van Kuik & Lignarolo

The convergence scheme of [23] (App. B.3) switches, after convergence of the Ψ -condition (damping 0.05), to the condition $v_n = 0$, applied as $\Delta r_i = -v_{n,i} d$ with $d = 0.0025$ (“fine tuning”). Transferring this to the midpoint-collocation code fails structurally: after Ψ -convergence, v_n at the first ≈ 5 rings is $O(1)$ (0.9 at the first ring) — a discretization artifact of midpoint collocation, not a usable residual — and stepping on it destroys the solution for either sign of the update. In the V4 discretization (Sec. 5) the artifact disappears ($v_n \approx 3 \cdot 10^{-2}$ at the edge), but the v_n -phase is then unnecessary, since the Ψ -iteration already converges to machine accuracy. Note that even in [23] the fine-tuning does not resolve the edge: the paper reports a 40% violation of $\Delta p_{sheet} = 0$ at the first ring and concludes that the boundary conditions hold for a vortex sheet but not for its leading edge.

number of rings (k)	error in c_P
4	$1.5 \cdot 10^{-4}$
8	$1.7 \cdot 10^{-5}$
16	$1.6 \cdot 10^{-6}$

Table 6: Error in c_P as a function of the number of rings

8 Summary and Discussion

This report describes an implementation of a vortex ring/cylinder model for investigating the constantly loaded axisymmetric actuator disk in the spirit of Øye, van Kuik, Bontempo and Manna. We confirm that the following model parameters influence its accuracy:

1. number of rings,
2. location of the semi-infinite vortex cylinder,
3. number of integration points at the disk,
4. model for the self-induction,
5. shape of the slip stream,
6. number and distribution of vortex rings close to the disk edge.

We therefore confirm the finding of Gijs van Kuik [22] that inter-relationships between the parameters of the model (see Table 1) exist. Nevertheless, an increase in the number of rings always increases the accuracy. To summarize:

1. Model accuracy: Table 6 gives an example of the accuracy reached (with respect to c_P from momentum theory) as a function of the number of rings (runs with $z_{SIVC} = 200$; the iteration was stopped when $\text{errc}_P < 10^{-6}$): In the light of Sec. 2.4, such numbers have to be read with care: since errc_P passes through zero crossings during the iteration, stopping on errc_P can flatter the accuracy. The honest single-run accuracy measure is the c_P fluctuation band reported by V3 of the code (e.g. $\pm 4 \cdot 10^{-4}$ at $NP = 600$, $z_{SIVC} = 25$); it decreases with NP and z_{SIVC} and can be reduced further by Richardson extrapolation in $1/NP^2$.

2. Convergence (new in Version 5): The sheet residual errsl and the momentum-theory deviation errc_P measure different things; errsl can be driven below 10^{-8} while errc_P stagnates at its discretization floor (Sec. 2.4). The V3 iteration ($\sqrt{z}$ -damping, optional Anderson acceleration) reaches the floor reliably within a few hundred iterations, where the original scheme needed 10^4 or never settled.
3. Edge singularity: Van Kuik [18, 22] and Chattot [7] proposed different edge behaviors ($\gamma \propto s^{-1/2}$ spiral vs. constant γ with logarithmic velocity singularity). Our converged solutions show a bounded γ with fitted exponent $a = 0.00 \pm 0.04$ — different from both, as expected for a single-valued, midpoint-regularized sheet discretization; see the detailed discussion in Sec. 7. Both v_z (Fig. 8) and v_r (Fig. 9) tend to finite values for $z \rightarrow 0^+$ in our (regularized) solutions.
4. Structural change (V4, new in Version 6): Node collocation with a spacing-proportional vortex kernel and a pinned first ring converges with uniform damping to a unique fixed point (residuals $\sim 10^{-12}$, no c_P band); at $N = 1600$ the c_P error is $+4.9 \cdot 10^{-5}$, already better than the V3 midpoint scheme at $N = 4000$. See Sec. 5.
5. Consistency: Do we fulfill Theorem 1? For errsl : yes (V3). For errc_P : only in the combined limit $NP \rightarrow \infty$, $z_{SIVC} \rightarrow \infty$ — at fixed discretization, errc_P has a floor no iteration can undercut.

9 Acknowledgments

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